

WHAT SETS THEM APART

HEIGHT	WEIGHT	WING SPAN
Up to 1.2 m	Up to 15 kg	Up to 2.5 m

- Normally fly 5-20 m high, but can reach 100 m for longer flights.
- Has a wide sideways vision to maximise predator detection, but a narrow frontal vision.
- Females usually deliver just one egg a year. The eggs are laid on the open ground, and face risks from feral animals.
- It is one of the heaviest flying birds in the world

ing any adverse effects on breeding in the wild, and shifting the eggs to the Sam hatchery.

It took the government five years to study breeding programmes and rope in international experts before signing a memorandum of understanding in 2018. The political wrangling around setting up a breeding centre was perhaps telling of the laxity among authorities. Initially, Kutch in Gujarat, with its moderate climate, mild temperature and higher rainfall being conducive to fertility, had been identified as the best site to set up a national breeding centre. But the state didn't have enough wild GIBs for a founding population, which meant that the only other source, Rajasthan, would have had to provide the eggs. But that didn't work out. Finally, scientists found an ideal location at Sorsan in Kota, but it took two years to finalise an MoU between Rajasthan, the Centre and the WII. A piece of land was allocated in Sorsan in 2019, but the project didn't take off as many didn't want mining activities in nearby areas to cease. So, a temporary hatchery was set up in Jaisalmer till the government established the permanent one at Ramdeora.

A BLEAK PICTURE

The captive breeding project finally seems to be making some headway after these bottlenecks, but the experience has been strenuous and the future of the GIB still hangs by a thread. "We are learning here with no precedents about artificial incubation, hatching, rearing and captive breeding," says an officer of the Rajasthan wildlife department. "The Houbara bustard has been reared and bred in captivity successfully. [But] we don't know how chicks born here will behave. GIBs are similar to Houbara, but different in weight, size and flight." For scientists, too, it has been a learning process, as captive breeding of GIBs is new. "These birds take about four years to mature. The oldest ones at our facility are three years old now, and it may take a few years before they start breeding in captivity," says Dutta.

Experts are also worried because the factors that pushed the GIB to this precipice remain. So, even a new generation born in captivity will be at risk once it is released into the wild. The major threats, as environmentalist and former bureaucrat M.K. Ranjitsinh points out, are power lines and solar parks. The GIBs, with poor frontal vision, often fail to detect power lines in time and get killed due to collisions. Almost 15 per cent of deaths occur because of high-tension lines, and the 13,000 sq km biodiversity-rich area around Jaisalmer city has a dense network of such transmission cables. Being terrestrial, the GIBs also walk more, and solar panels laid out across their desert habitat leave little place for them to forage and lay eggs. Birds that fly into Pakistan are often killed by hunters, while on Indian territory they face risks from animals.

A Supreme Court order in April 2021, following a public interest litigation filed by Ranjitsinh, directed that high-voltage transmission lines of all existing and new projects in Rajasthan and Gujarat, mostly solar and wind energy, be laid underground. The court also ordered that diverters be used to protect the birds. This has proved to be expensive—as per industry estimates, moving transmission lines underground may cost nearly Rs 22,000 crore—and thus, very little has changed on the ground. In fact, the Union ministries of environment, power and renewable energy had in the same year sought a modification of the order. Currently, the Centre and the Gujarat and Rajasthan governments are working together to file a plea stating that a low birth rate among GIBs is the key problem, not deaths due to power lines.

Conservation efforts have also encountered socio-economic hurdles. While it was easier to create protected areas for India's forest-dwelling tigers and lions, doing so on private land for GIBs, which live in agro-pastoral landscapes, has been difficult. For instance, state governments created protected areas for bustards in Solapur in Maharashtra and Karera in Madhya Pradesh in 1979 and 1981, respectively, but these encompassed large tracts of private land, thus affecting local livelihoods. The step alienated the very people who were earlier supportive of conservation efforts. Now, locals around the Desert National Park have been demanding that its area be reduced, and Gujarat officials have suggested that Rajasthan take their six female birds so that Gujarat, like Karera, which has been denotified, can free up some land. This has angered conservationists, who say that the native habitat of GIBs can be repopulated with birds born in captivity.

Amid these setbacks, the Great Indian Bustard continues to fight for its existence. In October, one was killed in Jaisalmer after it flew into a power cable. It was the second such death in the area since 2020. With their numbers dwindling, each death is a grim warning. Can the captive breeding project reverse the tide? Perhaps, but not without the support of civil society, the state, and a sincere commitment to development that's not at the cost of the environment. ■